STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Normal force per unit area.

Pa s.

A_1v_1=A_2v_2.

p+rhov^2+rhogh=constant.

N/m.

rhogh.

6pirv.

2S/r.

30,000 Pa.

Doubles.

Lower.

100 N.

40 Pa.

Bubble has two liquid-air surfaces; Deltap=4S/r.

8 m/s.

16,000 Pa.

approx 1.88x10^-4 N.

Hydrostatic balance gives p=p_0+rhogh due to weight of fluid above.

4.5 m/s.

Deltap=4S/r=60 Pa.